Machine Learning for Disaster Prediction



Goal

Create a model that predicts the **probability of survival of passengers** in a shipwreck

Dataset

	pclass	survived	name	sex	age	sibsp	parch	ticket	fare	cabin	embarked
0	1	1	Allen, Miss. Elisabeth Walton	female	29	0	0	24160	211.3375	B5	S
1	1	1	Allison, Master. Hudson Trevor	male	0.9167	1	2	113781	151.55	C22 C26	S
2	1	0	Allison, Miss. Helen Loraine	female	2	1	2	113781	151.55	C22 C26	S

Features
pclass: 1st, 2nd, 3rd class
name
sex
age
sibsp: total number of the passengers' siblings and spouse
parch: total number of the passengers' parents and children
ticket
fare
cabin
embarked: Port of Embarkation

Info

Rows: 1309
Columns: 13
Target column: **'survived'**
Not Survived: 0
Survived: 1

EDA

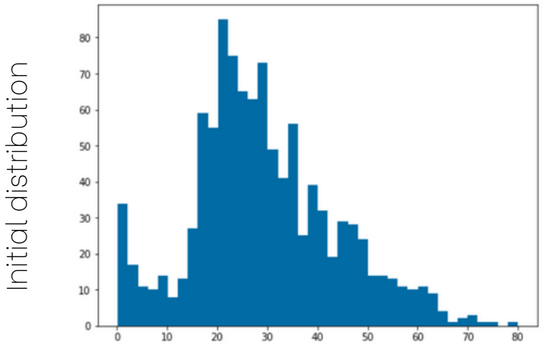


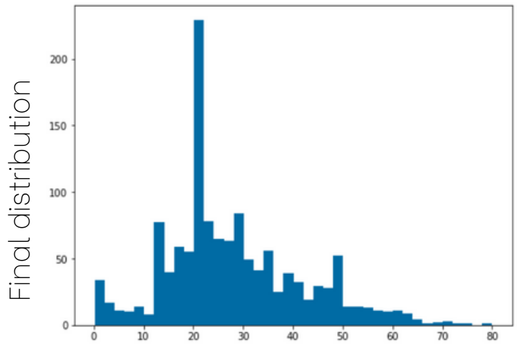
Missing Values (NaN)

age

Problem: 20% missing values

Solution: fill the missing values with random values centered around the mean and distributed with standard deviation





Result: most of the ages between 20 and 30 years (the average age of the passengers)

fare

Problem: only 1 missing value

Solution: comparison with similar passengers (sex/age/pclass)

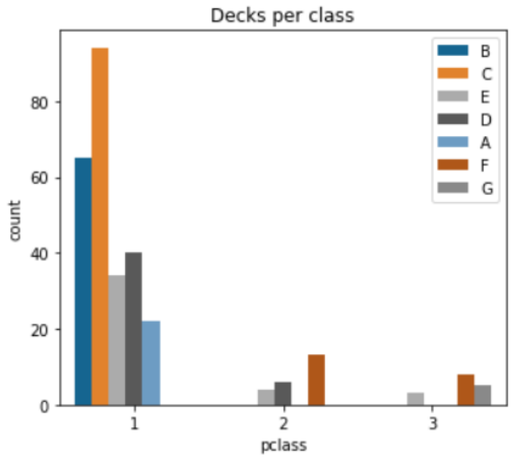
	pclass	name	sex	age	sibsp	parch	ticket	fare	cabin	embarked	survived
1225	3	Storey, Mr. Thomas	male	60.5	0	0	3701	NaN	?	S	0

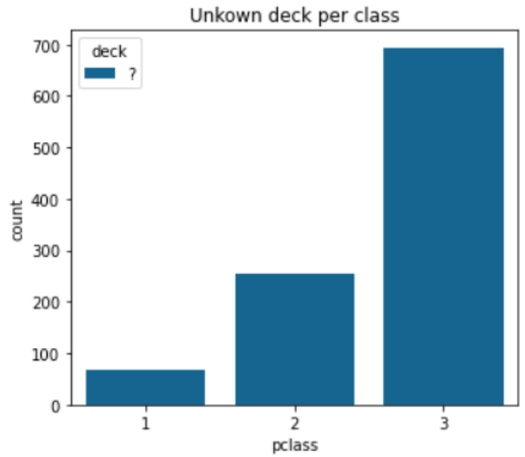
Unknown Values (?)

cabin

Problem: >70% unknown values

Solution: create a new column **deck** with the first letter of the cabin's code and fill unknown values randomly based on pclass





embarked

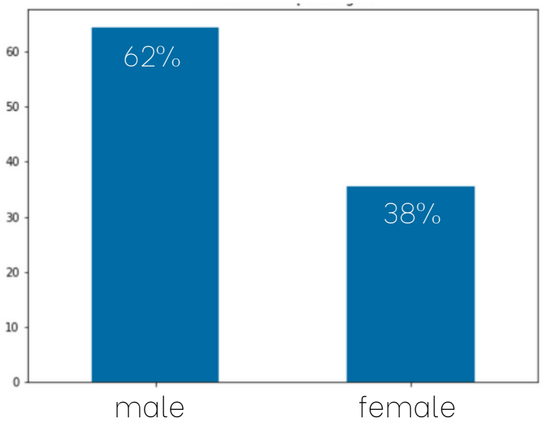
Problem: only 2 unknown values

Solution: fill values with the most common port of embarkation

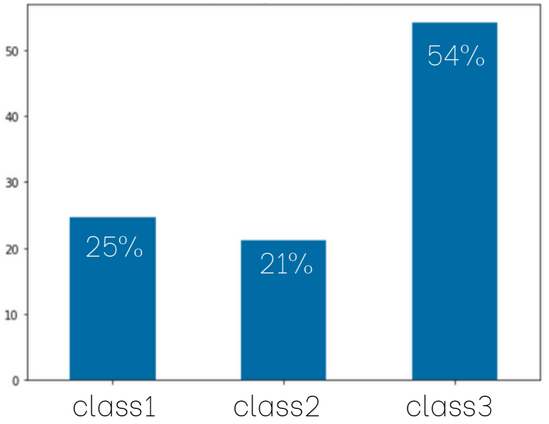
	pclass	name	sex	age	sibsp	parch	ticket	fare	cabin	embarked	survived	deck
168	1	Icard, Miss. Amelie	female	38.0	0	0	113572	80.0	B28	?	1	B
284	1	Stone, Mrs. George Nelson (Martha Evelyn)	female	62.0	0	0	113572	80.0	B28	?	1	B

Visualizations

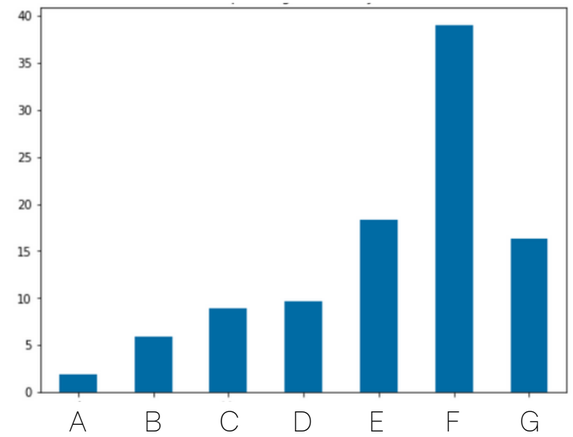
sex



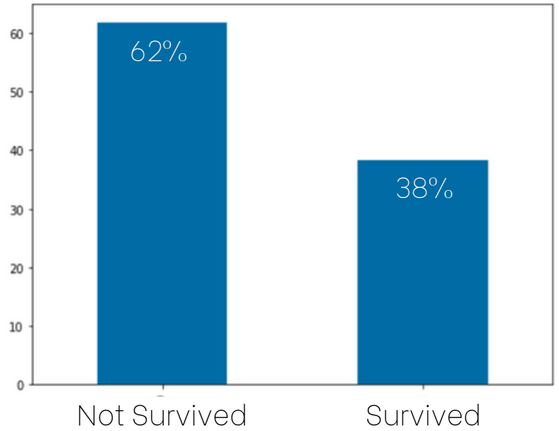
class



deck



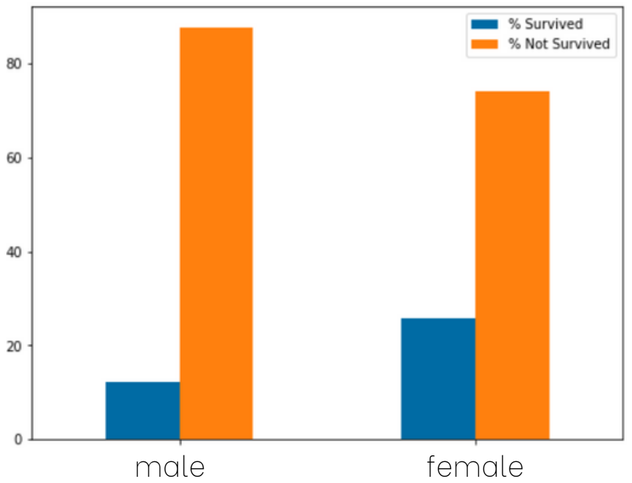
survived



More than **60% of the passengers have died in the shipwreck**.

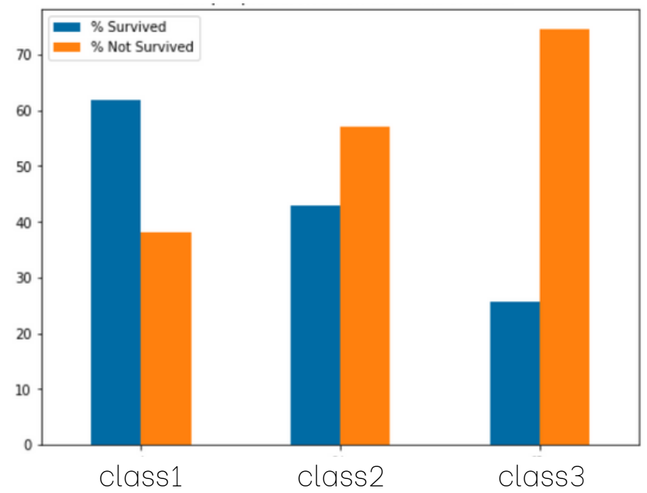
Visualizations

survivors/sex



More than 80% male passengers died as compared to around 70% female passengers.

survivors/class



More than 70% passengers of class 3 died as compared to around 60% passengers of 1st class survived.

feature engineering



Categories: New columns

name

Original: strings containing many different titles

[’Miss’, ’Master’, ’Mr’, ’Mrs’, ’Col’, ’Mme’, ’Dr’, ’Major’, ’Capt’, ’Lady’, ’Sir’, ’Mlle’, ’Dona’, ’Jonkheer’, ’Countess’, ’Don’,’Rev’, ’Ms’]

Transformation: creation of a new column ’**title**’ and replacement of the less known titles

[’Miss’, ’Master’, ’Mr’, ’Mrs’, ’Other’, ’Royal’]

sibsp/parch

Original: numbers that indicate families’ sizes

Transformation: creation of a new column ’**fam_size**’that summarize the numbers of family’s members [range(1,7)]

df[“fam_size“] = df[“sibsp“] + df[“parch“] + 1

Note: Group 7 contains families with 7-8-11 members.

New categories: Discretization

age

Original: passenger from 2 months to 80 years
Type: continuous

Transformation: dicretization (step = 10 years)

fare

Original: fare from 0$ to 512$
Type: continuous

Transformation: dicretization (step = variable)

Group 1: (0, 7.83]
Group 2: (7.83, 10.55]
Group 3: (10.55, 21.50]
Group 4: (21.50, 41.58]
Group 5: (41.58, 512.33]

Labeling

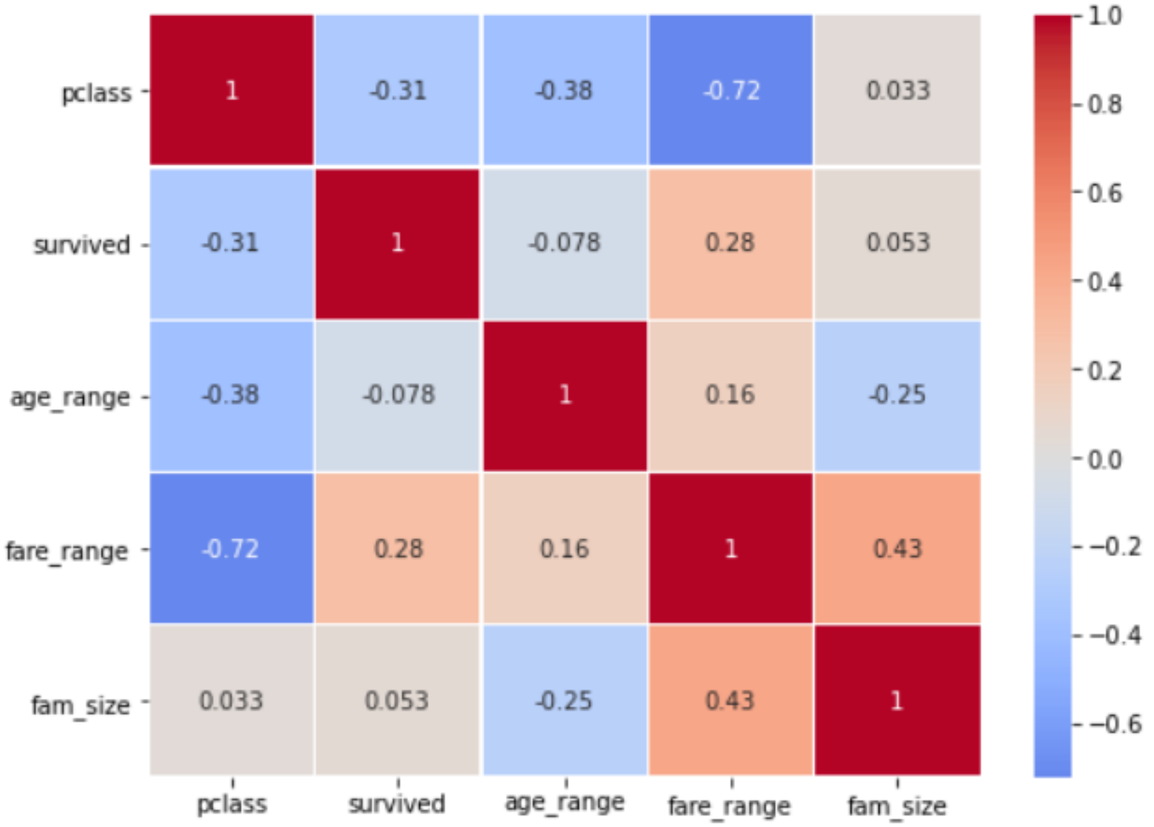
pclass: {pclass1 : 1, pclass2 : 2, pclass 3:3}
deck: {A : 0, B : 1, C : 2, D : 3, E : 4, F : 5, G : 6}
fare_range : {(0, 7.83]$: 0, (7.83, 10.55]$: 1, … , (41.58, 512.3]$:4}
sex: {male : 0, female : 1}
title: {Miss : 0, Mr : 1, Mrs : 3, Master :4, Royal :5, Other :6}
age_range: {[0:10]yrs : 0, [10:20]yrs : 1, … , [70:80]yrs :7}
fam_size: {1_person : 1, 2_people : 2, … , 7_people :7}
embarked: {S : 0, C : 1, Q : 2}
survived : {survived : 1, Not_survived :0}

	pclass	deck	fare_range	sex	title	age_range	fam_size	embarked	survived
0	1	1	4	1	0	2	1	0	1
1	1	2	4	0	3	0	4	0	1
2	1	2	4	1	0	0	4	0	0

Info

Rows: 1309
Columns: 8
Target column: **'survived'**

Correlation



No big collinearity... good data for ML model

best model



Model Selection

selection of classification algorithms

1. Logistic regression
2. Gaussian Naive Bayes
3. K nearest neighbor
4. Linear Support Vector Machine
5. Random Forest

split

Train/Test = 70/30 %

target (’survived’)

Not Survived: 0

Survived: 1

metrics

accuracy: true predicted values / total predicted values

and the winner is...

Random Forest > accuracy: **80,46%**

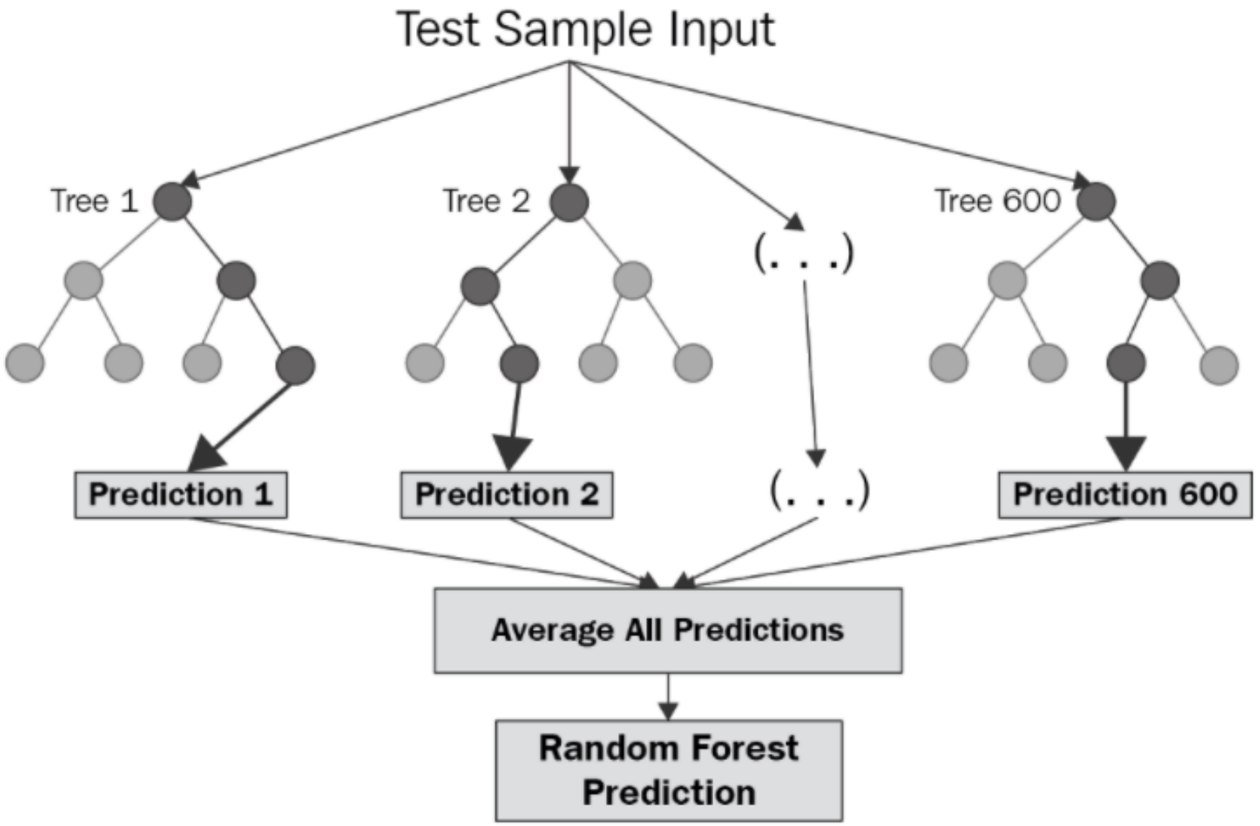
random forest



Random Forest

random forest

Supervised learning algorithm – **classification** problem
> **multiple random decision trees**.



Previous considerations

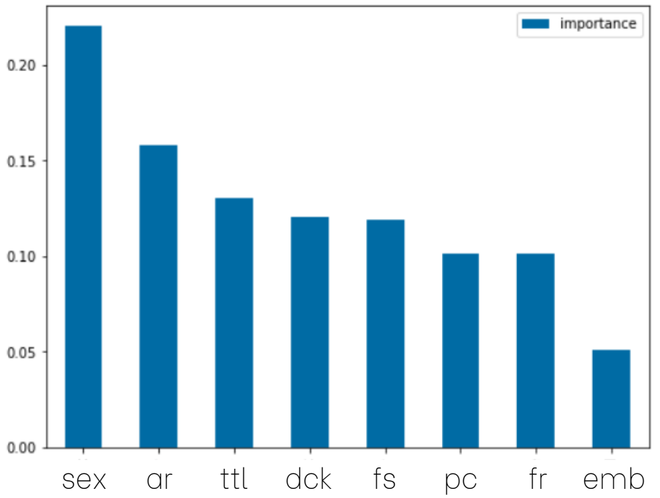
imbalanced target

Not Survived (0): 61.8%
Survived (1): 38.2%

Classes a little bit imbalanced

feature importance

Measures the contribution from every feature.



sex
ar: age range
ttl: title
dck: deck
fs: family size
pc: pclass
fr: fare range
emb: embarked

embarked doesn't play a significant role in our algorithm > we decide to remove it

Dataset Random Forest

split

Train = 70% data

Test = 30% data

	pclass	deck	fare_range	sex	title	age_range	fam_size	survived
0	1	1	4	1	0	2	1	1
1	1	2	4	0	3	0	4	1
2	1	2	4	1	0	0	4	0

Info

Rows: 1309

Columns: 7

Target column: **'survived'**

Not Survived: 0

Survived: 1

Model Tuning

grid search

Select the best hyperparameters for our model.

hyperparameters

- n_estimators : 300
- criterion: Gini
- max depth : 12

evaluation



Classification Report

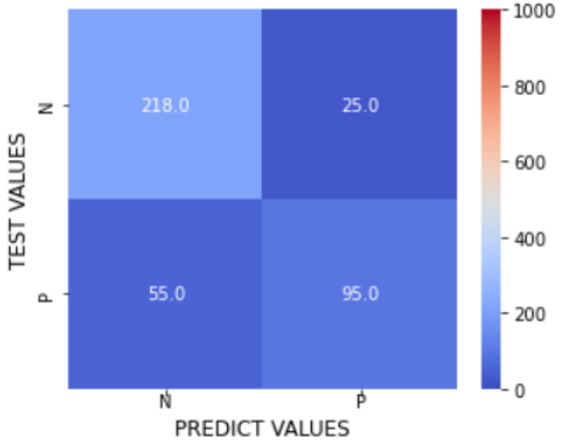
classification report

	precision	recall	f1-score	support
0	0.80	0.90	0.84	243
1	0.79	0.63	0.70	150
accuracy			0.80	393

The model **predicts 0's better than 1's**, probably due to the slight imbalance of the dataset.

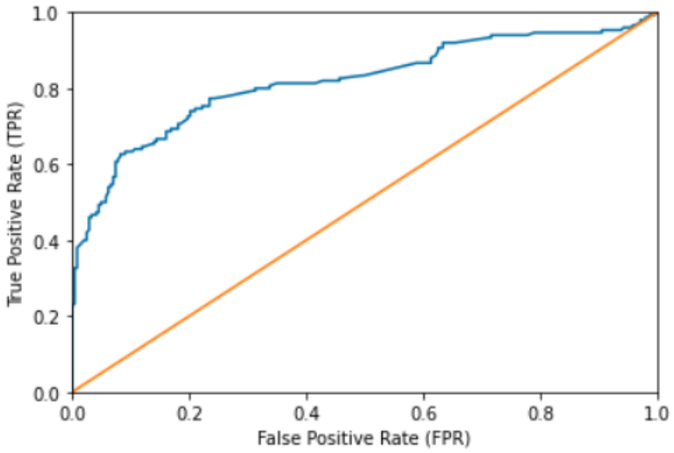
confusion matrix

TN: 218 (80%) **TP: 95** (79%)
FN: **55** (20%) **FP: 25** (21%)



ROC AUC curve

ROC AUC score: 82%



conclusions



Chances to Survive

~80% accuracy
~71% f1-score
~82% ROC-AUC score

The model is quite good in predicting the probability of survival of the passengers but there is still room for improvement.

Possible Improvements

1. extensive data exploration
2. the reduction of imbalanced data
3. deeper hyperparameter tuning

